

Scalars and Vectors

- Definition: A scalar is a single numerical value. It represents a magnitude or quantity and has no direction.

Ex: Force = 10N
↳ It is magnitude

⇒ Application in data Science:

- Like we have dataset of 3 features

x_1	x_2	x_3
-	-	-
-	-	-
-	-	-
-	-	-

Records

So, here the single value information we get is scalar like:

- Count of the Total no. of records = 4
- Avg of the feature x_1 = $\frac{\text{Sum of record value}}{4}$

(*) Simple Linear Regression:

$$y = mx + c$$

↓
Slope

↳ Intercept

↳ Scalar

[Not scalar because its value depend on x].

(2) Vectors: Numerical value which has both magnitude and direction.

A vector is an ordered list of numbers. It can represent a ^{data} point in space or quantity with magnitude or direction.

Ex: 10 N force in East direction.

Ex in terms of data science

- We have a dataset of fitness level in which we have 43 features.

Age	Weight	Height	Fitness Level
[18	60	170]	Fit
35	85	172	Average

So, these vector quantities represents the value in ordered way under features.

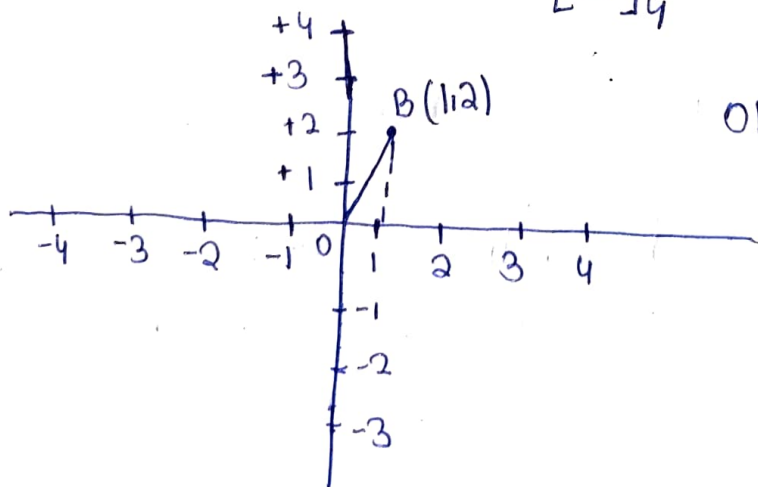
$[18, 60, 170] \checkmark$

$[60, 18, 170] \times$
(Unordered)

⊗ $\frac{60 \text{ kg}}{\text{Unit}}$
Magnitude

⇒ How data is presented in vector space.

Like we have data in vector. $A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_y^x \Rightarrow 2 \text{ dimension}$

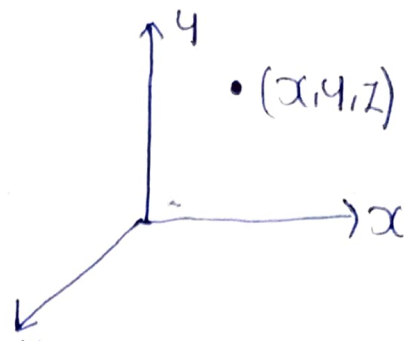


$$OB = \sqrt{5}$$

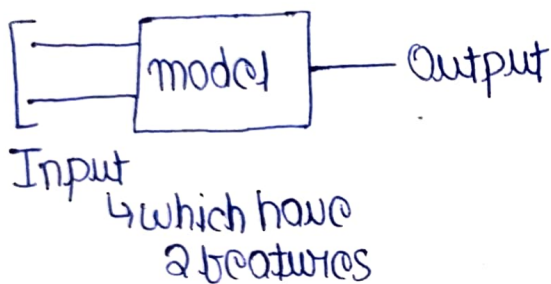
$$\text{Hyp} = \sqrt{1+4} \\ = \sqrt{5}$$

It 3-dimension:

$$B = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$



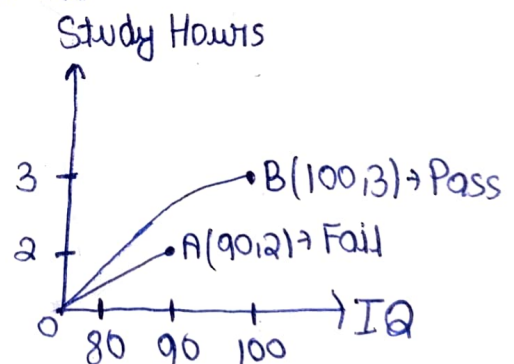
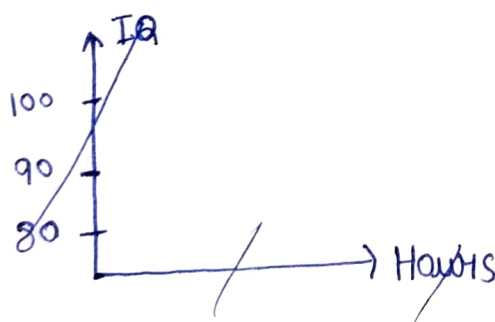
- Every data point of the dataset can be represented in vector form.
- Here, we will see how these things really help to train models.



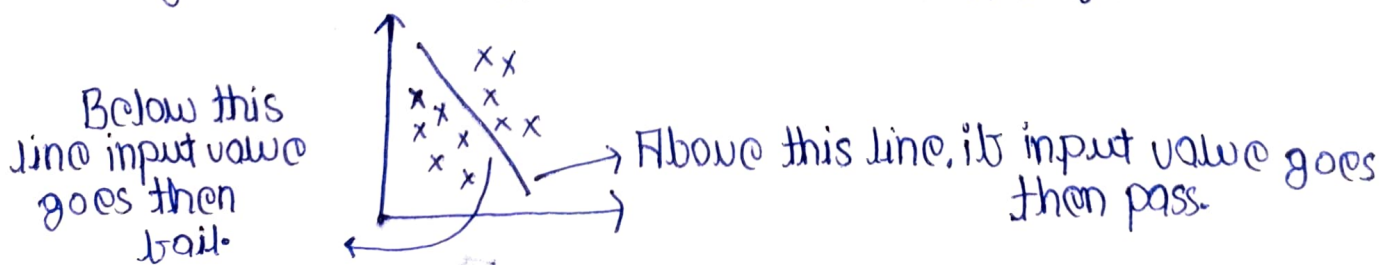
* We trained a model on large dataset which tell whether a student will pass or fail based on his IQ and study hours.

IQ	Study hours	Result
90	2	Fail
100	3	Pass

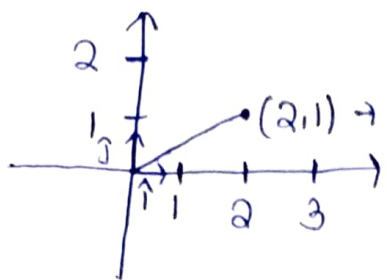
So first each data point will be presented as:



⇒ So, by this when more data points are given and then some ML algorithms are used whole data is now classified.



- Unit vector is a vector whose value is 1.



Unit vector = 1

$\hat{i}$ and $\hat{j}$ are unit vectors towards x and y axis and whose magnitude is 1.

- * Vectors are also majorly used in Gaming Industry.

Like we played racing games, and when 2 cars/object collide there were some effects like car catches fire, slows down, or get rotated in air.

In all these vectors are responsible.

- * Vectors can be anywhere in the co-ordinate system with same different origin point but in same direction and equal length.

